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Reservoir Formation Fluid Equilibration & the Complexity of Contact Identification

B. Elshurafa and M. Berkane, Aramco, Saudi Arabia; A. M. Ali and M. Charupa, SLB, Saudi Arabia

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Abstract

Accurate hydrocarbon column identification is becoming increasingly challenging as we encounter more complex plays, both in terms of lithology and pore structure. In these environments, direct movable fluid measurement along with advanced interpretation techniques is essential for reducing uncertainty regarding fluid contacts and enhancing our understanding of reservoir fluid equilibrium. Gradient analysis applications are best utilized in intervals with good porosity, single phase, and relatively thick zones. The methods proposed in this paper utilize direct fluid measurement to confirm movable fluids, thereby overcoming the limitations of the conventional approach. A calibrated resistivity measurement is deployed to provide direct salinity calculations even in the presence of oil. The adopted workflow integrates real-time data to reconstruct saturation modeling and facilitate better identification of various fluid contacts. Finally, results were plotted with access pressure to indicate changes. A new reservoir fluid geodynamics technique was also used to confirm oil complexities beyond pressure gradient answers; such changes have little effect on gradient analysis.

Results presented in this paper show an improved understanding of reservoir fluid variations. In a 150 ft thick interval, water and oil movement were identified throughout the layer. A direct fluid resistivity measurement was critical to this observation, confirming that the reservoir did not have enough time to equilibrate and establish a sharp oil-water contact. The gravitational forces were not sufficient to overcome capillary forces holding water high in the structure. Traditional gradient analysis confirms the presence of a gas-water contact, while the direct fluid measurement identified a small oil rim between the gas and the water. The gradient uncertainties window typically masks the small oil rim by increasing the gas density. PVT analysis confirmed fluid changes within the small interval. In the absence of gradient analysis, reservoir fluid geodynamics was utilized to confirm oil property variation with depth. The optical spectroscopy measurement indicated oil density increasing vertically during the real-time data gathering, results were later confirmed by laboratory PVT analysis. A novel residual pressure gradient analysis based on fluid density was used to anchor the pressure gradient trend to resolve the presence of two different fluids, result was confirmed by downhole fluid identification using optical spectroscopy measurement and later by laboratory analysis. Accurate direct downhole resistivity measurements were instrumental in reducing the unknowns

in the Archie equation. The application of reservoir fluid geodynamics identified oil property variation to confirm reservoir filling and equilibration complexity real time

Introduction To Reservoir Fluid Equilibrium

Geologically oil or gas accumulation in reservoir is expected to reach an equilibrium, density differences enable gravitational forces to segregate the two phases vertically if sufficient time is given. Such process can be hindered by baffles or later reservoir movements. In the case oil-water reservoir, Free Water Level (FWL) is established. A part from equilibrium, which is always assumed, pressure gradient analysis has several uncertainties that should be accounted for. These are pressure gauges accuracy, temperature effects, capillary pressure effects and fluid compositional gradient see [Dumont et al 2014 \[1\]](#), a workflow that determine free water level is presented by [Dubost et al 2015 \[2\]](#). Dynamic reservoir characterization data gathering is critical to field development and management; therefore, it is important to gather formation testing flow tests to confirmed and validate saturation models. Preparation of these data often take longer time using conventional approaches. Innovative approaches that seek solutions beyond normal formation testing is presented in this paper. Due to the geological settings and fast area of activities, evaluation challenges vary significantly and are not confined to a particular one reservoir or location. Accurate reservoir evaluation requires an accurate fluid movement. Wells drilled with water-based mud (WBM) are difficult when quantifying the water fraction in the presence of water-based mud filtrate (WBMf) if accurate fluid resistivity cannot be measured downhole.

Carbonate reservoirs have a complex pore structure, heterogeneity, and varying mineral composition that leads to a difficult fluid distribution evaluation using inversion-based saturation modelling that heavily rely on resistivity logs and formation water salinity. When the density difference is small between oil and water, the capillary forces keep water saturation high up in the structure and might often not have a sharp contact and a long transition prevail and introduce significant error to the evaluation.

Deterministic approach to saturation and contact evaluation

Pressure gradient analysis is a crucial technique used in the oil and gas industry for reservoir characterization and fluid dynamics assessment. It involves measuring pressure at various depths within a well using downhole gauges or sensors and plotting these measurements against depth to create a pressure gradient profile. The slope of this gradient can reveal the type of fluid present—such as oil, gas, or water—due to their differing densities. By identifying changes in the gradient slope, fluid contacts like the gas-oil contact (GOC) or FWL can be inferred, which are essential for understanding reservoir structure and estimating recoverable hydrocarbons. This analysis not only aids in differentiating fluid types but also helps assess reservoir connectivity, as variations in pressure gradients between wells may suggest compartmentalization. Furthermore, pressure gradient analysis assists in recovery planning by optimizing the water injector placement to manage reservoir pressure. However, the effectiveness of pressure gradient may be limited in scenarios involving highly compressible fluids, such as light hydrocarbon in which fluid density change with depth while gradient analysis assume constant. Reservoirs with complex textures and variable wettability also introduce complexity in contact determination as in it is also influenced by drilling fluid type oil-base or water base mud. Reservoirs heterogeneity may lead to varying porosity and permeability, and as result affect the pressure response which is typically seen on producing assets. The reliability of this analysis is contingent upon high-quality data collection, necessitating precise sensors and careful calibration. Despite these challenges, pressure gradient analysis remains a vital tool for petroleum engineers and geoscientists, offering critical insights into reservoir dynamics and facilitating efficient hydrocarbon resource management.

In scenarios where robust saturation modelling is not feasible due to low resistivity contrast, low porosity or complex porous space structure and other factors, wireline formation testing emerges as the

predominant technique for fluid identification using downhole fluid analysis technique to compliment petrophysical modelling. While reservoir fluid contact estimation through pressure gradient analysis performs optimally in conventional reservoirs, its application in complex reservoirs is hindered by challenges such as supercharging effects, which make obtaining valid reservoir pressure difficult or nearly impossible. Downhole Fluid Analysis (DFA) offers a solution by bridging the gap in evaluation when pressure gradient data is unreliable. The initial generation of DFA tools focused on identifying water and oil while monitoring oil-based mud (OBM) contamination in real-time, as noted in references [3], [4], and [5]. Advancements in DFA technology have introduced new sensors that provide quantitative analysis of fluid composition, including components such as C1, C2, C3, C4, C5, C6+, CO₂, gas-oil ratio (GOR), density, and Fluorescence, which are crucial for establishing reservoir connectivity [6]. Recently a calibrated direct fluid resistivity measurement was introduced with a new wireline formation testing platform. The platform enables real time water resistivity measurement to differentiate water fluid downhole. The principal resistivity measurement is based on induction. Details of the measurement principle is presented by Pfeiffer et al 2020 [7]. Recently Manuaba et al 2025 [8] presented the application of down resistivity sensor to direct determine reconstruct saturation models and identify of sweet spots.

The application of DFA to design perforation is well-established Chu and Welch [9] and Chu and Steckhan [10], [11] presented solutions in wells with complex saturation modelling due to abnormal resistivity response due to conductive minerals. In these wells gradient analysis was not possible due to limited valid pressure point and small reservoir thickness that increase gradient uncertainties. When reservoirs quality is poor, inflatable packers or 3D radial probe were utilized to extends the operational envelope, enabling fluid analysis even in lower permeability reservoirs as low as 0.05 md/cP mobility, as demonstrated by Ali et al. [12]. These advancements underscore the versatility and critical importance of DFA in overcoming the limitations of traditional pressure gradient analysis, thereby enhancing reservoir evaluation and informing strategic decisions in exploration and production. Ali et al [12] 2020 presented the challenges of formation testing in carbonate reservoir showing that over 1754 stations evaluated in this study are indicating that 68% of these stations are tight reservoir exhibiting significant challenges in evaluation. This study confirms the value of increase inlet to flow-area is an enabler for reservoir fluid characterization. On the other hand, gradient analysis was not the best approach to develop these intervals.

Examples of Oil-Water Equilibrium

Figure 1 shows the pressure vs depth plot of vertical well that encounter thick 50ft potential oil. Gradient analysis based on the pressure measurements indicates oil with 0.87g/cc density a typical black oil, a water leg was also identified by gradient analysis implying a high salinity water based on high water density of 1.15g/cc. Formation testing was deployed to gather fluid sampling to evaluate both water and oil. Formation testing tools with calibrated resistivity measurement & optical spectroscopy measurements were deployed to characterize water and hydrocarbon while recovering fluid samples. The oil layer was identified to have movable water throughout the structure with increasing trend as we move up the oil column as shown in Fig 1(a). Water salinity was calculated downhole using fluid conductivity measurement and was found to be 165kppm consistent with high gradient density. Since the well was drilled with water base mud, the obtained results illustrate continuous transition extending over 50ft vertical thickness. Similarly, Fig 1(b) is another well across the same reservoir and it exhibit a higher movable water up the structure, it is worth nothing this well is downdip.

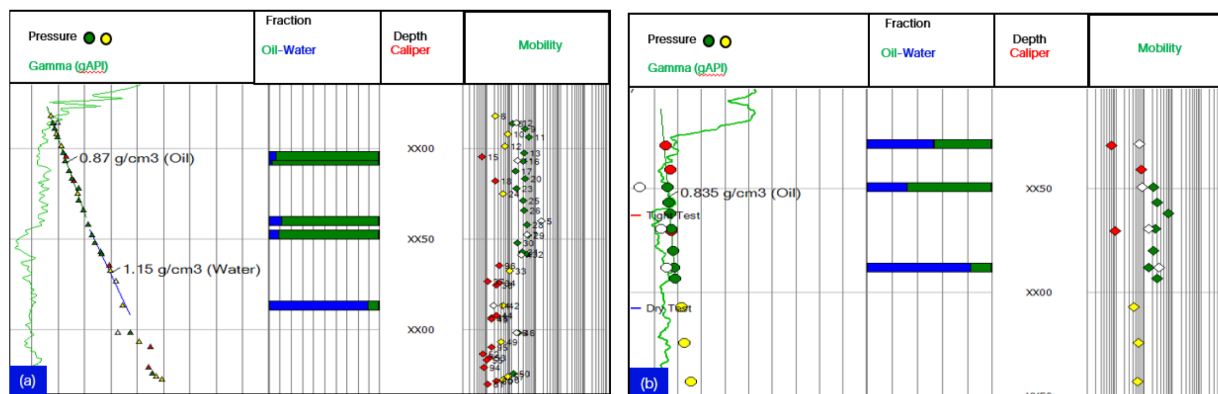


Figure 1—The pressure vs depth plot of two vertical wells: (a) well with the oil layer containing movable water identified with downhole fluid analysis, (b) another well across the same reservoir showing higher amount of movable water.

Shown in Fig 2 is the image log interpretation, the analogue wells AX12 is about 50ft up dip however exhibits a similar characteristic with regards to water movement. The bedding for the AX03 is different from AX12 consistent with highly compartmentalized reservoir. More data is required to confirm if there are other phenomenon that can explain oil-water movements.

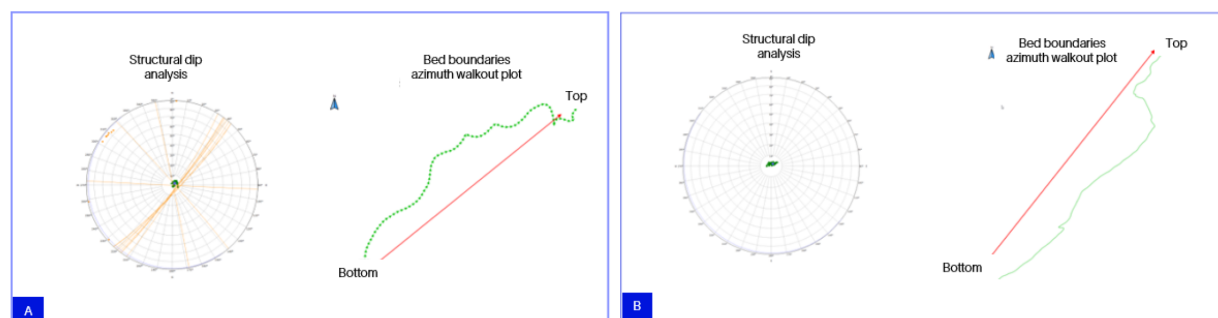


Figure 2—The structural dip analysis is using bed boundary dips as input. Well (ax03) section is 8.38in showing one structural dip zone with gentle structure towards NE (53.9degE/2.8 deg) while well (bx12) is also 8.38in shows one structural dip zone with sub horizontal structure with inclination of 2.4 deg.

The cross-section correlation illustrated in Figure 3 presents a detailed examination of the complex dynamics within the reservoir, offering critical insights to fluid behavior and guiding future exploration and production strategies. This analysis highlights notable differences in reservoir behavior across wells. Recent drilling operations have successfully intersected oil reservoir in two additional wells after a previously discovered oil. This discrepancy underscores the dynamic nature of reservoir characteristics and the importance of continuous analysis. Pressure data comparison reveals a lack of hydraulic connectivity among the three wells, as evidenced by a total pressure difference of approximately 150 psi in True Vertical Depth Subsea (TVDSS) measurements. Such pressure variations are pivotal in understanding fluid flow and reservoir segmentation. The movement of fluids within the reservoir suggests that the formation has not had sufficient time to equilibrate, preventing the natural segregation of water and oil based on their density differences. The presence of water at the top of the oil column is particularly intriguing, posing questions about fluid dynamics within the reservoir. This anomaly is compounded by the fact that the wells were not drilled using water-based mud, indicating that any water movement originates solely from the reservoir itself. Among the wells, one exhibits a more pronounced presence of water at the top of the structure, in contrast to the other two wells, which show only 3 to 10% water movement. This variability highlights the complex interplay of reservoir parameters and fluid interactions.

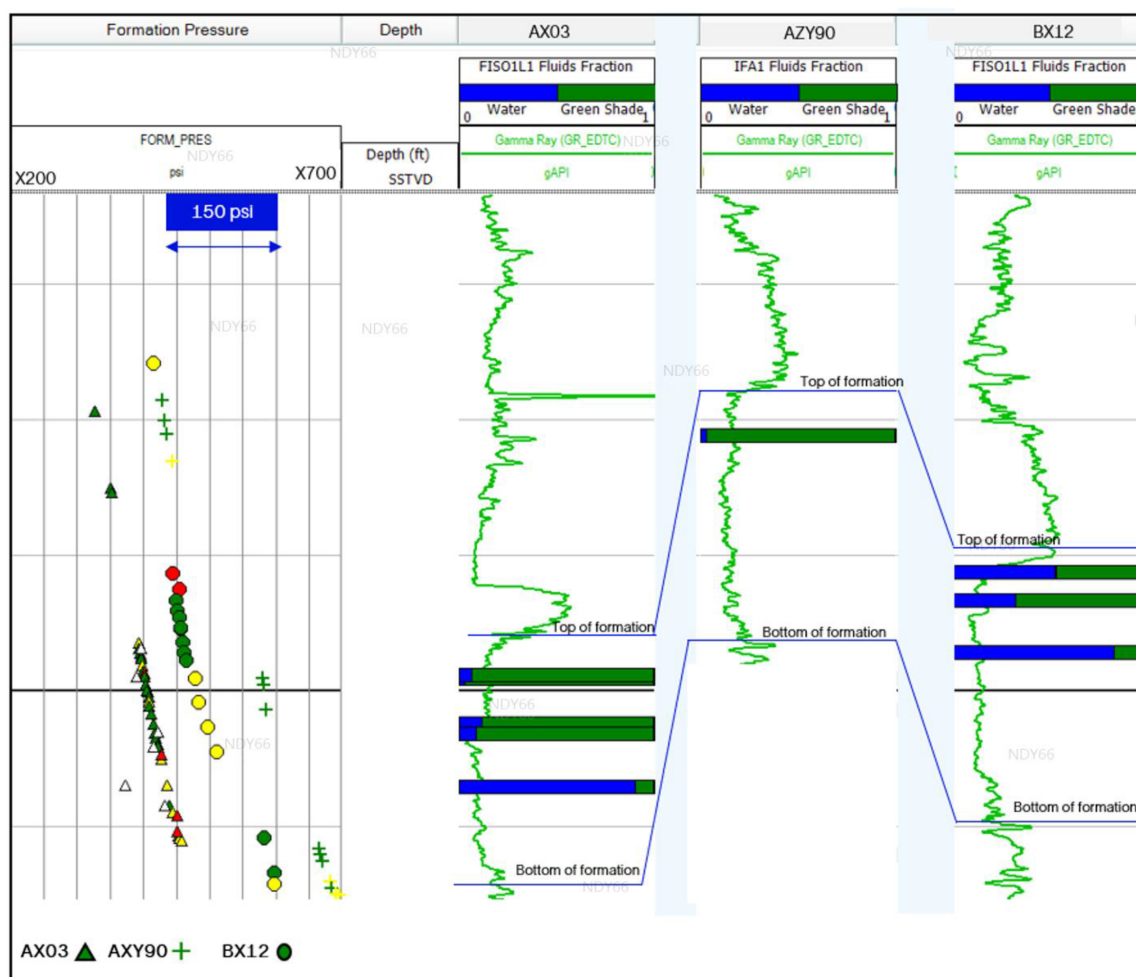


Figure 3—Complexity of contact evaluation, the effect of water in thick tight carbonate reservoir, effect of capillary pressures due to tightness(a) Part 1; (b) Part 2; (c) Part 3. 3% water fraction was observed on well AZY90 which is about 180ft up dip. The characteristics of this well is consistent with well AX03. While BX12 is 60ft up dip and has much higher moveable water.

Pressure gradient analyses further illuminate the dominant fluid dynamics, indicating that oil is the predominant fluid with a density of 0.87 g/cc, while water displays a density of approximately 1.15 g/cc. These values are derived from meticulous gradient analysis, providing a foundation for understanding fluid distribution. Downhole fluid property measurements, conducted using a new formation testing tool, reveal a water density of 1.13 g/cc and salinity levels of 161kppm. Concurrently, oil flows with a density of 0.89 g/cc, consistent with its dominant presence in the reservoir. These measurements are crucial for accurately characterizing the reservoir and informing extraction strategies. The complexity of fluid saturation and movement within the reservoir can be attributed to several key factors: poor rock quality, capillary pressure effects, low density contrast between water and oil, and the impacts of drilling operations. These elements collectively contribute to the observed fluid behavior, complicating reservoir characterization and management strategies. Understanding and accounting for these factors is essential for accurate reservoir evaluation and the development of effective management approaches.

Based on these results presented, there is no common free water level or consistent fluid contact due to reservoir compartmentalization. It is expected to have different fluid equilibrations based on the region in discussion. There are also other processes that can alter contacts and free water level such as tilting known as hydro-dynamism.

A comprehensive flow test conducted subsequently further confirmed the intricate nature of fluid saturation within the reservoir. Formation testing and sampling have provided an exhaustive characterization

of the reservoir, revealing detailed information about fluid properties and rock permeability, obtained from multiple build-up tests. The observed results can be attributed to capillary pressure that retain higher water fraction on the top of the reservoir. In summary, the insights gained from these analyses are critical for refining reservoir characterization and optimizing resource management.

Fluid fraction

Figure 4 offers a comprehensive examination of downhole fluid identification, serving as a crucial component in the analysis of reservoir dynamics and fluid properties. This detailed investigation is essential for optimizing extraction strategies and enhancing the accuracy of reservoir characterization. Panel (a) presents the pressure and rate history, which begins with an attempt at fast pump-out. However, this effort was impeded by the reservoir's low permeability, resulting in a rapid decline in flowing pressure. Such challenges underscore the importance of understanding reservoir characteristics early in the evaluation process. In response, the flow rate was adjusted to establish a steady state, ensuring more stable conditions for ongoing analysis and extraction.

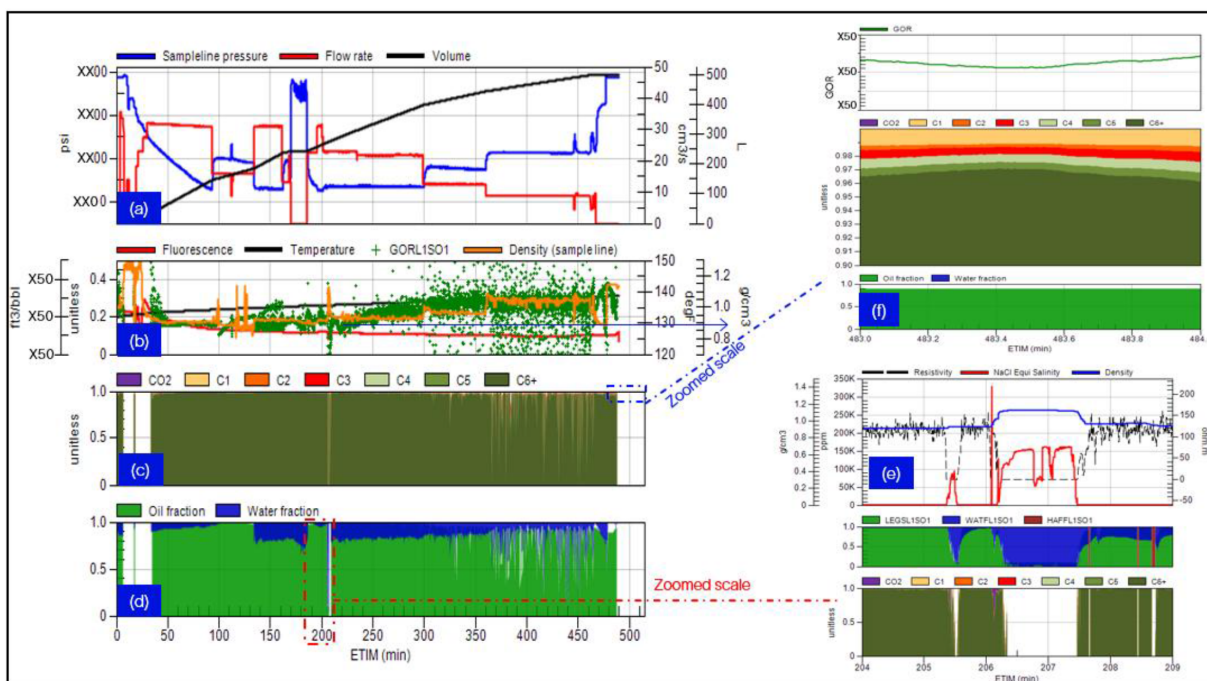


Figure 4—An example of downhole analysis in the presence of two-phase flow – oil and water. (a) shows the downhole pressure, flow rate and cumulative volume, (b) fluorescence, temperature, GOR and fluid density, (c) hydrocarbon composition, mainly represented by heavier C6+ fractions, (d) fluid fractions, where blue shading is water, green is oil, (f) shows a zoom on the composition of an oil slug, which shows that C6+ almost 96%, (e) shows a zoom on the water slug with salinity 165kppm.

In panel (b), fluid resistivity and density were meticulously measured using two independent sensors. This dual-sensor approach provides a robust framework for capturing accurate data on fluid properties, minimizing errors and enhancing reliability. Complementing these measurements, optical spectroscopy with the channels distributed across 400 to 2100 nm range, was employed for real-time fluid analysis, offering insights into fluid fractions, hydrocarbon composition, Gas-Oil Ratio (GOR), and CO₂ content. This sophisticated analysis revealed that the flow was predominantly oil, with approximately a 15% water fraction, as detailed across Figure 4.

The fluid flow characteristics were identified as either slug flow or mixed flow. Slug flow is particularly beneficial for obtaining precise fluid property measurements, given its ability to capture discrete packets of fluid with distinct characteristics. In contrast, mixed flow can lead to the overestimation or underestimation

of fluid properties, posing challenges for accurate assessment. The zoomed plots in Figures 4e and 4f provide detailed insights into specific aspects of fluid characterization, focusing on water and hydrocarbon properties, respectively. Further enhancing the analysis, a calibrated induction resistivity measurement indicated a water salinity of 165kppm and a density of 1.13 g/cc, flowing concurrently with oil. This measurement is critical for understanding the interaction between different fluid phases and their impact on reservoir behaviour. Figure 4f highlights oil with a low GOR, predominantly composed of C6+ hydrocarbons, with minimal light ends. This composition is significant for assessing the quality and marketability of the extracted hydrocarbons.

Overall, the analyses presented in Figure 4 offer a detailed understanding of fluid properties and composition within the reservoir. Such insights are invaluable for accurate reservoir characterization, enabling informed decision-making and strategic planning. By leveraging detailed measurements and advanced analytical techniques, stakeholders can optimize resource management, improve extraction efficiency, and enhance production strategies. This comprehensive approach underscores the importance of integrating technology and expertise in the pursuit of effective reservoir management and exploitation.

Gradient reconstruction using downhole fluid analysis. In the early stages of oil and gas exploration, static open-hole logs serve as a vital tool for evaluating reservoir saturation. Despite their utility, these logs can introduce significant uncertainties due to the limited availability of geological and petrophysical data at this phase. To mitigate these uncertainties, initial evaluations often rely on assumptions derived from the accessible data to provide a foundational understanding of the reservoir. One of the key methodologies employed in this context is pressure gradient analysis, which helps in identifying reservoir fluid contacts. Fig.5 shows the pre-assumed hypotheses concerning saturation modelling to derive critical points such as the Gas-Water Contact (GWC). The ability to accurately determine these contacts is essential for understanding the distribution of fluids within the reservoir, which in turn influences production strategies and resource estimation. The illustrative example highlights a potential GWC determined through a two-fluid theory. During the fluid sampling process, comprehensive downhole fluid analysis confirmed the presence of water, oil, and gas within the reservoir. Interestingly, the interval initially indicating oil presence was masked by the gas gradient, thereby validating the existence of a GWC. The excess pressure calculation shows a reliable gas gradient and indicate change of fluid consistent with two fluid theory.

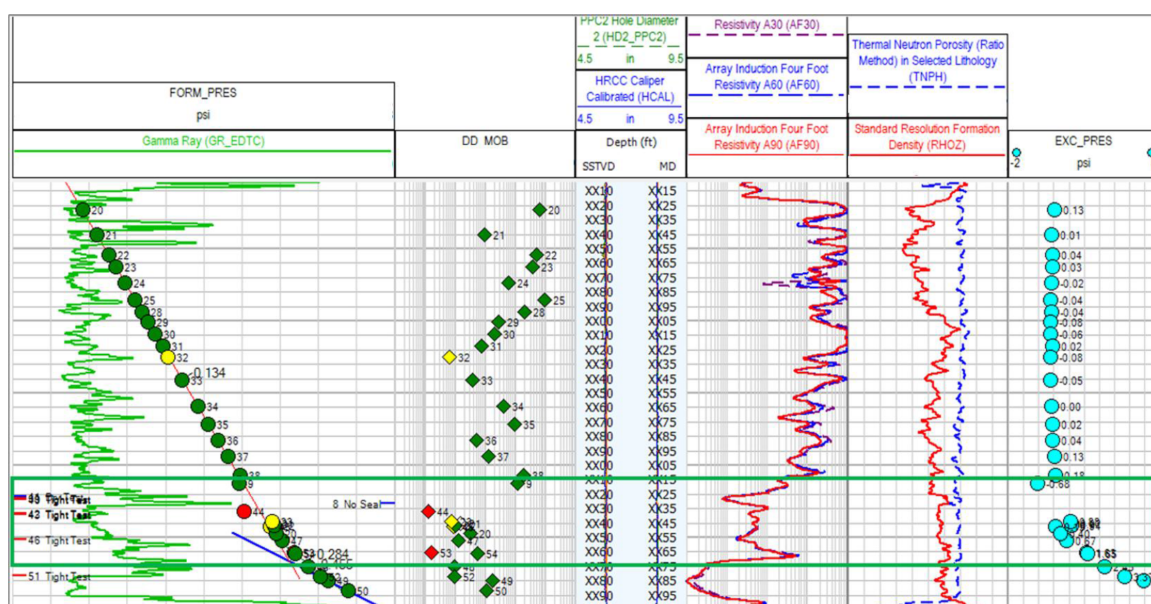


Figure 5—Complexity of fluid contacts evaluation with pressure gradients. The presence of oil is masked by gas and water gradients.

To enhance precision in fluid contact determination, a series of downhole fluid analysis stations were employed as shown in Fig.6. This figure shows fluid properties in different tracks, hydrocarbon composition is shown in (b), fluid fluorescence (d), fluid density (e) and gas oil ratio (f). This meticulous approach allowed for accurate pinpointing of the oil down-to and gas down-to levels. This detailed analysis led to the discovery of an additional 15 feet of light oil situated near saturation pressure.

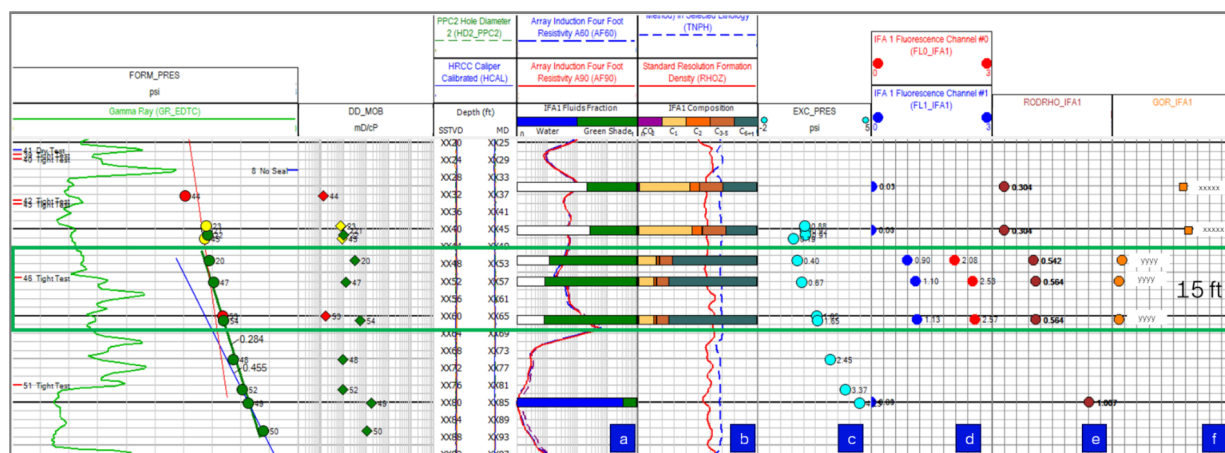


Figure 6—Uncertainties in direct gradient contact evaluation. The first two tracks show pressure and pressure gradient analysis. Takes (a) to (f) show the integration of downhole fluid analysis to reduce contact determination. Tracks (a) show the hydrocarbon and water fractions while (b) represent hydrocarbon composition. Tracks (d), (e) & (f) are fluorescence, fluid density & gas oil ratio respectively gathered during sampling and downhole fluid analysis.

Such findings are significant, as they can impact the overall assessment of the reservoir's potential and guide future development and production plans. The identification of light oil near saturation pressure also suggests potential challenges and opportunities in managing reservoir pressure and maximizing recovery. The contrast measurement with highest contrast in the hydrocarbon measurement is GOR 1648 scf/bbl in oil compared to 18812cf/bbl in the gas leg. The fluorescence and fluid density also show a large difference to confirm the hydrocarbon fluid change from gas to oil. Finally, the methane content is dominant in the gas column as well. The use of static open-hole logs combined with pressure gradient analysis and downhole fluid analysis provides a robust framework for reservoir characterization during exploration. These methodologies are essential for refining the understanding of fluid distribution and improving the accuracy of fluid contact determinations. As exploration progresses, these insights contribute to more informed decision-making, optimized resource management, and strategic planning for development and production.

Figure 7 presents a detailed analysis of downhole fluid identification, integral to assessing reservoir dynamics and fluid properties for optimized extraction strategies. Track (a) represents the pressure and flow rate history indicating an excellent reservoir permeability with minimum drawdown. Flowrate was reduced to avoid going below saturation pressure to secure a quality formation fluid sample. Real time data-driven adjustments are crucial for enhancing the precision of reservoir characterization. Track (b) shows gas oil ratio and viscosity of oil pump, the increase GOR trend is an excellent indicator of clean-up from oil-based mud filtrate (OBMf) to a formation oil. Track (c) presents downhole fluid density and fluorescence the transition of OBMf to formation oil. While chart (d) shows the fluid composition at start of the pump, we observe only C6+ which is mainly mud filtrate while at full clean-up we observe the stabilization of C1, C2, C3-C5 stabilization which is generally absent from OBM composition. When a full stabilization is achieved on all the fluid properties one can proceed with sample capture.

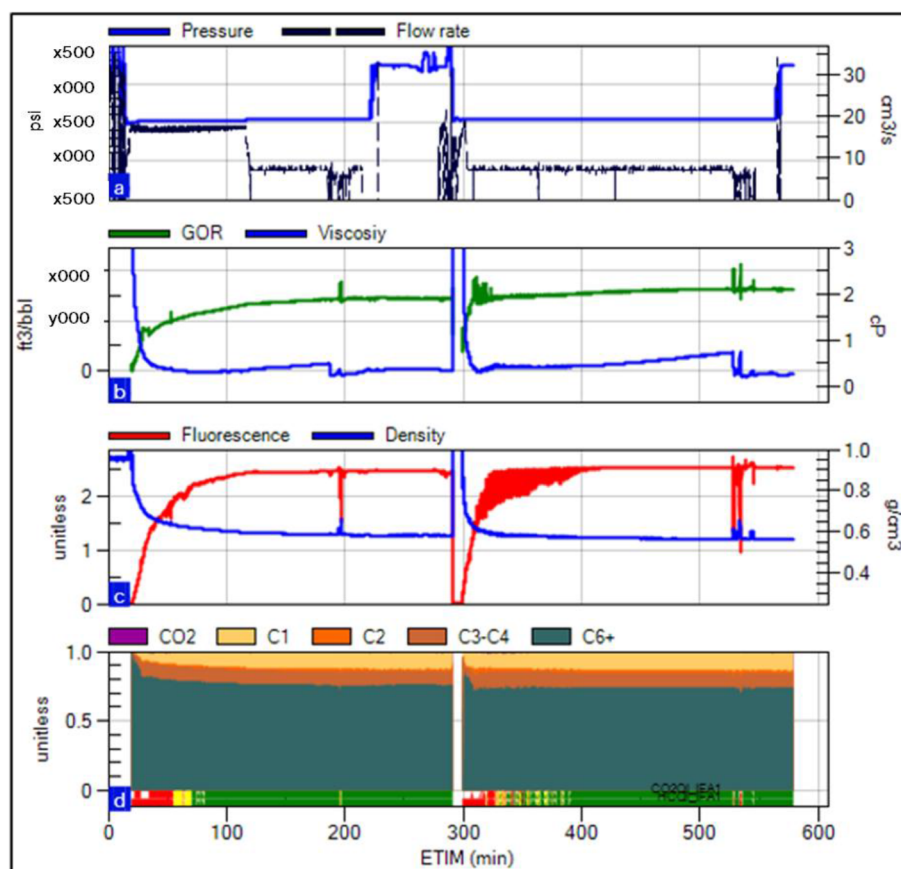


Figure 7—Downhole fluid analysis example. (a) represents the pressure flow rate history, (b) shows the gas oil ratio and viscosity measurement, c is density and fluorescence while (d) completes the hydrocarbon composition. As time increases stabilization is obtained on all measurements confirming a clean oil sample with minimum oil base mud filtrate contamination.

Deterministic approach to contact delineation. In the realm of oil and gas reservoir characterization, gradient analysis remains a fundamental tool for elucidating fluid distribution and establishing fluid contacts. Despite its widespread use, practitioners must be aware of its limitations, particularly in scenarios involving highly compressible fluids or reservoirs exhibiting complex textures and grain sorting. Gradient analysis is optimally suited for classical high-porosity and relatively thick reservoirs, where its underlying assumptions regarding fluid behaviour are more likely to be valid. Figure 8 presents an analysis that identifies a light hydrocarbon with a density of 0.51 g/cc gradient constructed across a 50-foot interval, in the lower reservoir fresh water with a density of 1 g/cc was identified. This analysis enabled the establishment of a contact point between the water and the light hydrocarbon at a depth of XX67 feet, a crucial interval for comprehending fluid interactions and distributions within the reservoir.

At multiple depths oil and water samples are recovered with minimum drilling mud contamination. Final analyses are obtained from laboratory using gas chromatography to derive hydrocarbon composition and Pressure-Volume-Temperature (PVT) measurements that provided fluid insights. Water ion analysis is also performed while also salinity is inverted from conductivity measurements.

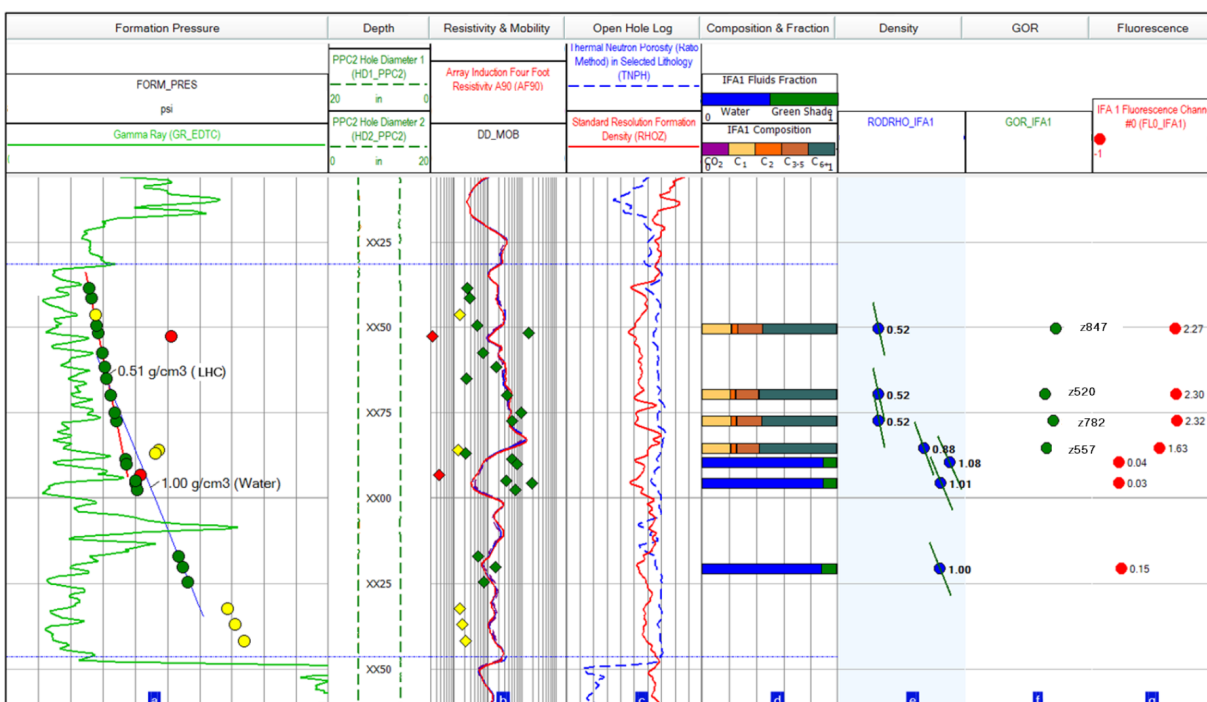


Figure 8—Complexity of contact evaluation, the effect of water gradient landing in hydrocarbon zone. (a) show the Gamma Ray pressure and pressure gradient. (b) show the resistivity and mobility, c shows the density and porosity log, d presents the composition of hydrocarbon and water fractions. While (e), (f) & (g) are fluid density, gas oil ratio and fluorescence respectively gathered during sampling and downhole fluid analysis.

Downhole fluid analysis assisted contact

Direct fluid density measurements conducted at multiple depths using DFA demonstrated remarkable consistency with gradient-driven density values, yielding a measurement of 0.52 g/cc compared to 0.51 g/cc a variance of merely 2%. Similarly, comparisons of water density between gradient analysis and direct downhole fluid measurements confirmed a density of 1.0 g/cc, corroborated by open-hole logs that exhibited a slight resistivity response, as depicted in Figure 8, track (b). It is also worth noting potential two water layers with different density that is critical to the contact re-construction.

The hydrocarbon fluid identified suggests a more liquids than gas as inferred from high fluorescence, elevated density, a minimal methane fraction, and a lower gas-oil ratio indicative of volatile oil or condensate. It is worth noting that gas typically does not emit fluorescence. In the absence of reliable water gradients below the hydrocarbon zone, the measured fluid density is utilised to construct a gradient and establish logical water contact based on direct flow measurement. A density of 1.08 g/cc was used to plot the gradient trendline intersecting the depths indicated by downhole fluid analysis, as shown by the dotted gradient trendline in Figure 8. This analysis led to the identification of a new contact at XX86 feet, approximately 19 feet deeper than the initial direct contact evaluation. These findings underscore the inherent uncertainties associated with relying solely on gradient contacts utilization for fluid contact evaluation, which introduces greater uncertainties in the fluid data.

Such meticulous analytical efforts are crucial for developing an accurate and comprehensive characterization of the reservoir, thereby informing strategic decisions related to reservoir management and production optimization. By integrating advanced analytical methodologies and recognizing the limitations inherent in gradient analysis, industry professionals can achieve a more nuanced understanding of the reservoir. This, in turn, facilitates efficient and sustainable resource extraction and management, ensuring optimal production strategy.

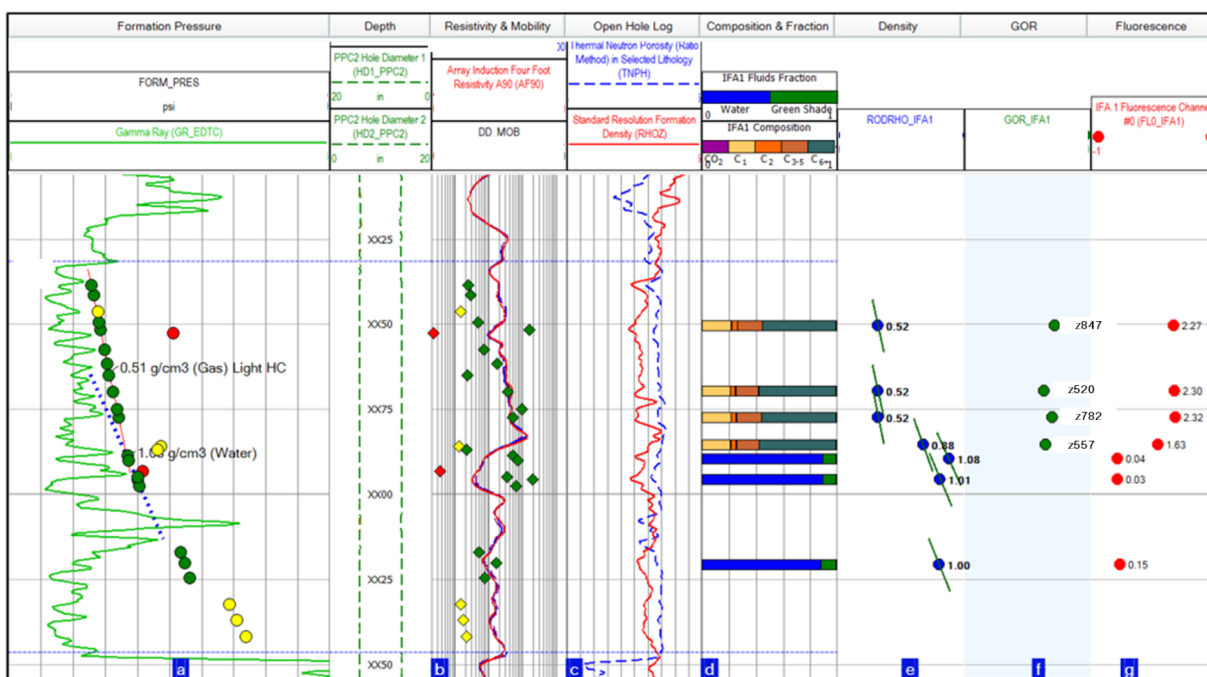


Figure 9—Contact evaluation, the effect of water gradient landing in hydrocarbon zone. (a) show the Gamma Ray pressure and pressure gradient. (b) show the resistivity and mobility, (c) shows the density and porosity log, (d) presents the composition of hydrocarbon and water fractions. While (e), (f) & (g) are fluid density, gas oil ratio and fluorescence respectively these are gathered during sampling and downhole fluid analysis.

Conclusion

The new method of direct fluid identification and characterization with downhole fluid analysis described in this paper has been proven to be the most reliable and effective for saturation identification. It identifies fluid contacts and variation in saturations in the most complex cases presented above, where traditional methods, such as pressure gradient analysis, often overlook the data: undersaturated low permeability carbonates with a mixed saturation of oil and water, sandstone reservoir with a thin oil rim located between gas and water, a heterogeneous oil reservoir with a water below divided into two zones with different pressure regimes.

This advancement significantly reduces uncertainties related to the overestimation or underestimation of resources, providing a more accurate assessment of reservoir potential and fluid dynamics. Furthermore, integrating these methods with reservoir fluid geodynamics could provide deeper insights into regional geological processes and their impact on reservoir filling and fluid equilibration, enhancing our understanding of the complex interactions within the reservoir. This leads to more efficient extraction techniques and improved management of the reservoir, ultimately maximizing recovery rates and resource utilization ensuring that extraction processes are tailored to the actual conditions within the reservoir.

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